ANALYSIS REPORT

RECURRENCE RISK PREDICTION FOR THYROID CANCER ACCORDING TO ATA 2025 GUIDELINES

Retrospective Study, 114 patients

Study period: 2024

Data encoded according to ATA 2025 standards

1. Study Sample Characteristics

Total patients: 114. Recurrence events: 7 (6.1%).

1.1. ATA 2025 Risk Stratification

(ATA 2025 PTC 4-level classification)

Risk Level	N	%	Recurrence
Low	69	60.5%	5/69 (7.2%)
Low-Intermediate	2	1.8%	1/2 (50.0%)
Intermediate-High	15	13.2%	0/15 (0.0%)
High	28	24.6%	1/28 (3.6%)
Total	114	100%	7/114 (6.1%)

1.2. Descriptive Statistics of Study Variables

Variable	N (n=114)	%
Multifocal (>1 nodule)	31	27.2%
Bilateral involvement	32	28.1%
Tumor size >1cm	21	18.4%
Tumor size >4cm	1	0.9%
T stage T3+	28	24.6%
LN metastasis (N+)	59	51.8%
Lateral neck meta. (N1b)	8	7.0%
Central LN >=5	16	14.0%
Lateral neck LN (+)	21	18.4%
BRAF V600E	45	39.5%
Other BRAF	2	1.8%
TERT mutation	6	5.3%
TP53 mutation	2	1.8%
Any mutation	51	44.7%
Co-mutation (>=2 genes)	4	3.5%
BRAF + TERT	3	2.6%
BRAF + TP53	0	0.0%

2. Univariate Analysis Results

2.1. Univariate Logistic Regression

The table below presents univariate logistic regression results for each variable against recurrence.

Variable	N	OR	95% CI	p-value
Multifocal (>1 nodule)	114	2.116	0.446 - 10.048	0.346
Bilateral involvement	114	2.017	0.425 - 9.566	0.377
Tumor size >1cm	114	0.725	0.083 - 6.363	0.772
T stage T3+	114	0.494	0.057 - 4.288	0.522
LN metastasis (N+)	114	6.113	0.712 - 52.515	0.099
Lateral neck meta. (N1b)	114	2.381	0.251 - 22.622	0.450
Central LN >=5	114	2.657	0.469 - 15.039	0.269
Lateral neck LN (+)	114	0.725	0.083 - 6.363	0.772
BRAF V600E	114	1.161	0.247 - 5.449	0.850
BRAF (any)	114	1.074	0.229 - 5.038	0.928
TERT mutation	114	<0.001	--	1.000
TP53 mutation	114	<0.001	--	1.000
Co-mutation (>=2)	114	<0.001	--	1.000
BRAF + TERT	114	<0.001	--	0.999

Note: TERT, TP53, co-mutation and BRAF+TERT have non-estimable OR (complete separation) due to zero recurrence events in mutation groups. OR <0.001 indicates complete separation and has no real statistical meaning.

2.2. Key Observations from Univariate Analysis

- Lymph node metastasis (N+) was the strongest predictor of recurrence (OR=6.11; p=0.099), approaching but not reaching statistical significance at the 0.05 level.
- Multifocality (OR=2.12), bilaterality (OR=2.02), and central LN >=5 (OR=2.66) showed a trend toward increased recurrence risk.
- BRAF V600E showed OR=1.16, not statistically significant (p=0.85).
- TERT and TP53 had very few mutations in the sample (6 and 2 patients) with zero recurrence events, causing complete separation.

3. Multivariate Analysis Results

Multivariate logistic regression with 4 different models.

3.1. Model 1 - Clinical variables (N=114, events=7)

Variable	OR	95% CI	p-value
Size >1cm	0.528	0.058 - 4.797	0.570
T3+	0.415	0.047 - 3.701	0.431
N+	7.027	0.804 - 61.429	0.078

3.2. Model 2 - Genetic variables (N=114, events=7)

Variable	OR	95% CI	p-value
BRAF V600E	1.144	0.243 - 5.384	0.865
TERT	<0.001	--	1.000

3.3. Model 3 - Combined clinical + genetic (N=114, events=7)

Variable	OR	95% CI	p-value
Size >1cm	0.511	0.056 - 4.671	0.552
T3+	0.393	0.041 - 3.798	0.420
N+	6.933	0.786 - 61.193	0.081
BRAF V600E	1.070	0.206 - 5.569	0.936
TERT	<0.001	--	1.000

3.4. Model 4 - N+ and co-mutation (N=114, events=7)

Variable	OR	95% CI	p-value
N+	5.885	0.684 - 50.617	0.107
Co-mutation	<0.001	--	1.000

4. Key Findings and Conclusions

- **Recurrence Rate**

  The overall recurrence rate was 6.1% (7/114 patients), consistent with studies on papillary thyroid cancer patients receiving curative treatment.

- **ATA 2025 PTC Stratification**

  ATA 2025 PTC 4-level risk distribution: Low (60.5%), Low-Intermediate (1.8%), Intermediate-High (13.2%), and High (24.6%). The 4-level classification replaces older 3-level systems. However, due to incomplete data (missing histology, margins, ETE, vascular invasion), recurrence rates did not increase monotonically (Low: 7.2%, Int-High: 0%, High: 3.6%), highlighting the need for complete histopathological assessment.

- **Lymph Node Metastasis (N+)**

  N+ was the most important independent predictor, consistently identified in both univariate (OR=6.11; p=0.099) and multivariate analysis (OR=7.03; p=0.078). This was the strongest trend even though statistical significance was not reached due to small sample size.

- **Multifocality and Bilaterality**

  Multifocal (OR=2.12) and bilateral (OR=2.02) tumors showed a trend toward increased recurrence risk, consistent with ATA guidelines for intermediate-risk classification.

- **BRAF V600E**

  BRAF V600E mutation rate was 39.5% but showed no significant association with recurrence in this study (OR=1.16; p=0.85).

- **Rare Mutations**

  TERT (5.3%), TP53 (1.8%), and co-mutation (3.5%) had low frequencies. Complete separation occurred due to zero recurrence events in mutation groups, reflecting the small sample size and rare events.

- **Limitations**

  The limited number of recurrence events (n=7) substantially reduced statistical power, producing wide confidence intervals. Larger studies are needed to validate these associations, particularly for rare genetic variables. Survival analysis (Cox regression) may be more informative with longitudinal follow-up data.

Data encoded from Q3 source file and genetic results file. Variables coded according to ATA 2025 standards. Encoded data file: DATA_MAHOA_ATA2025.xlsx